



Probability Distributions

Key terms

discrete random variable, continuous random variable, probability distribution, probability density function, binomial distribution, normal distribution, continuity correction

KSBs: K13, S11, S52, S53, B1

3.1 Probability Distributions

Suppose we roll a fair 6-sided die *sixty times* and record the number of ones, twos, etc.

If we went on rolling the die, we might expect *roughly equal proportions of each number* to come up on the die. In saying this, you would be using a perfectly reasonable *model for how the probability is distributed.*

# Rolled	Frequency
1	12
2	9
3	11
4	10
5	7
6	11

We could express this model as:

x	1	2	3	4	5	6
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

If we want to consider the case $x=1$ (the die shows a 1)

which means "the probability that the die shows a 1." This table shows the probabilities of all its possible values is *equal to 1*. We write:

we can write $P(X=1)$ probability distribution of X .

3.2 Discrete Random Variables

Definition 3.1 - Random Variable

A **random variable** (r.v.), usually denoted X is a variable (a symbol which takes one of a range of values) where the sum of the probabilities of all its possible values is *equal to 1*. We write:

$$\sum P(X=x) = 1$$

The range of values a variable can take is called the sample space.

Example 3.1. In the example given above, X is the random variable "the number shown on the die"

Definition 3.2 - Discrete Random Variables

If a random variable can take one of a finite set of values we say it is a discrete random variable.

Example 3.2. Consider the situation where we roll two fair six-sided dice. Let X be the random variable

"the sum of the numbers on the two dice".

The sample space here is the whole numbers between 2 and 12. The probability distribution of X is given by:

x	2	3	4	5	6	7	8	9	10	11	12
$P(X = x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

For example, rolling a sum of 6 can be achieved in 5 ways (15, 24, 33, 42, 51) so it has a probability of $\frac{5}{36}$ and we say $P(X=6) = \frac{5}{36}$. We have $\sum P(X=x) = 1$ as $1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1 = 36$.

Exercise 3.1. (a) Let X be the denomination of coins put into a car parking meter. It is known that X follows the following probability distribution:

x	1p	2p	5p	10p	20p	50p	£1	£2
$P(X = x)$	0.02	0.02	0.05	0.07	0.08	0.16	0.4	0.2

- (i) What is $P(X \geq 10p)$?
- (ii) What is $P(5p < X \leq 50p)$?
- (iii) What is $P(X \leq 25p)$?

(b) A random variable has the following distribution.

x	1	2	3	4
$P(X = x)$	42%	18%	14%	?

What is $P(X = 4)$?

3.2.1 Mean of a Discrete Random Variable**Definition 3.3 - Mean of a Discrete Random Variable**

For a discrete random variable X , the mean of X , denoted $E(X)$ or μ can be calculated as

$$E(X) = \mu = \sum x P(X=x)$$

For each of the values x that X can take, we multiply x by the probability that $X=x$, and then take the sum. The mean is also referred to as the expectation/expected value of the random variable.

Example 3.3. For the 'sum of two dice' example above, the expected value is:

$$E(X) = \sum x P(X=x)$$

$$= \frac{2+6+12+20+30+42+40+36+30+22+12}{36} = \frac{252}{36} = 7$$

Note

If the values a discrete variable can take are *not numerical* (e.g., letter die, coin flip) we still need $\sum P(X=x) = 1$ but we *cannot compute an expected value*. We will mainly be working with variables that take numerical values.

Exercise 3.2. (a) The discrete random variable X has the following probability distribution:

x	-100	-50	-20	-10	0	100	200
$P(X=x)$	0.19	0.05	0.24	0.01	0.1	0.001	0.409

- (i) Find the probability that the random variable takes a negative value.
(ii) Find the expectation of X .
(b) For the two probability distributions given in Exercise 3.1, find the mean of each.

3.2.2 Variance of a Discrete Random Variable

We can also calculate the variance of a discrete random variable.

Definition 3.4 - Variance and Standard Deviation of a Discrete Random Variable

For a discrete random variable X , the **variance** of X is given by

$$\sigma^2(X) = \sum (x - \mu)^2 P(X=x) = \sum x^2 P(X=x) - \mu^2$$

The **standard deviation** $\sigma(X)$ of a discrete random variable is the square root of the variance.

Note that here, instead of dividing by n or $n-1$, we use *a probability* (which is already a proportion out of n). Since the probability distribution gives us information about the whole population, this calculation corresponds to the one used in finding *the population variance*.

Exercise 3.3. For the probability distributions given in Exercise 3.1 and 3.2, find the variance.

3.3 Continuous random variables

As well as defining probability distributions for discrete random variables, we can also consider continuous variables - which can take any value within a range - and describe how the probability changes as the value changes. In this case we call it a probability density function rather than a probability distribution.

Definition 3.5 - Continuous Random Variable

A continuous random variable is one which can take any value within a range.

Definition 3.6 - Probability Density Function

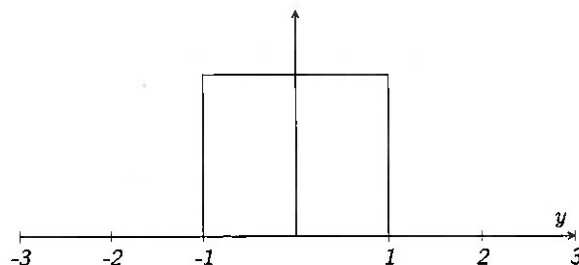
For a continuous random variable, a probability density function is a function that describes how the probability of the variable taking a given value is distributed.

For continuous random variables, the probability of the variable taking any one specific value is zero. The value of the probability density function at a given point in the sample space can be interpreted as a relative likelihood that the value of the random variable would be equal to that value - for example, if it is twice as high, the likelihood of that value is twice as much. (It's not a graph showing what the probability is, but the density of the probability, which can be thought of as where the probability is.)

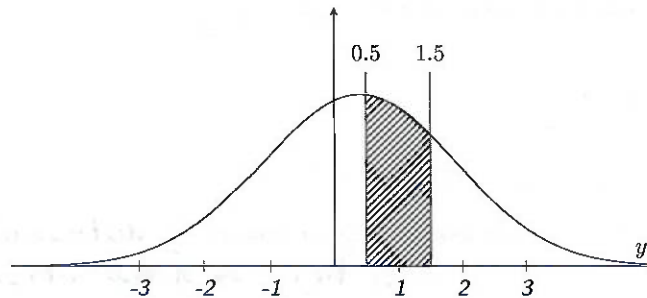
While we can't use the function to find the probability of a particular value, we can instead use it to find the probability of the variable taking values in a given interval. This can be calculated by finding the area under the graph between two points. We will see how this works later.

The area under the whole curve will always be 1 since this is the total probability of the distribution.

Example 3.4. The plot below shows a density function with uniform probability for which all values in the interval $(-1, 1)$ occur with equal probability. Since the width of the rectangle is 2 we know the height must be $\frac{1}{2}$ since the area under the curve is 1.



This plot shows a **non-uniform probability distribution** where values around 0.4 are the most likely and the probability gets smaller as you go away from this on either side. The region of values between $X=0.5$ and $X=1.5$ are shaded, and the area gives $P(0.5 < X < 1.5)$

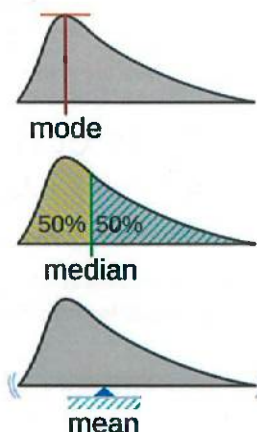


It is also possible to plot the **cumulative probability distribution** for which at each point the height of the curve gives the area under the original curve to the left of that point. The cumulative probability curve increases from left to right and has a maximum value of 1.



We can also extend the definitions of statistics from discrete to continuous random variables, and they have reasonably intuitive geometric interpretations:

- The mode of a continuous random variable is the value of X at the highest point of the curve
- The median of a continuous random variable is the value of X with 50% of the area under the curve to either side.
- The mean of a continuous random variable is the value of X which represents the "centre of mass" of the curve
 - if you were to make a physical model, it would balance on that point.



In practice, the only probability density functions you'll ever really need to work with are ones whose shapes match specific known mathematical functions, which we will see some examples of below.

If you have a set of data to work from, you don't need a probability density function, as you can instead *analyse the data*. - and in general, we won't be able to create a probability density function of any shape we want. What we can do is *find a curve that approximates the data* or *models the kind of real-world processes we are studying* and use that to make predictions.

3.4 Binomial Distribution

Definition 3.7 - Binomial Distribution

The **binomial distribution** is a *discrete probability distribution* which measures how likely it is that, if we perform *n trials of a two-outcome process* which succeeds with probability *p* we have *K successes*. If a random variable *X* has a binomial distribution, we can write

$$X \sim B(n, p) \text{ and seek to find } P(X = K)$$

Note

We use the symbol $\sim$ to mean "*has the distribution*"

There are four criteria we need for a situation to be modelled using a binomial distribution:

- There are only *Two possible outcomes to each trial (success/failure)*
- Each trial is *Independent of any other trial* (i.e. the outcome of one trial doesn't affect any other trial)
- There is a fixed *Number of trials in the experiment*
- The probability of *Success is always the same*.

Example 3.5. Suppose I flipped a fair coin 5 times and measured the number of times it lands on heads.

- There are *two outcomes (heads = success, tails = failure)*
- Each trial is *independent*
- There are a fixed *number of trials: n = 5*
- The probability of *success is always the same; There is always a 50% chance of heads for each flip.*

The four conditions are satisfied, so a binomial distribution can be used.

So if *X* is the number of times the coin lands on heads, then $X \sim B(5, 0.5)$

Exercise 3.4.

- (a) For each of the situations below, state which of the conditions of being modelled using a binomial distribution it fails to satisfy.
- (i) A coin is flipped until it lands "tails".
 - (ii) An archery student fires 13 arrows, and the number of times they hit the target is recorded.
 - (iii) A group of 15 friends arrive at the college, and the number of students who get to class late is recorded.
 - (iv) A die is rolled and the score on the die is recorded.
- (b) Write down the expression for the binomial distribution for each of the examples below, by determining the values of n and p in each case.
- (i) In a class of 30 students, it is expected that 22% of them will have brown eyes.
 - (ii) In one week, each day from Monday to Friday, Noel catches a train from Piccadilly station. He claims that the probability that the train arrives on time at the station is 0.4, and that the train's arrival time is independent from day to day.

3.4.1 Finding probabilities using a binomial distribution

Example 3.6. A fair coin is flipped 5 times, and let X be 'the number of heads'. So $X \sim B(5, 0.5)$
What is $P(X = 3)$?

There are 10 ways to get exactly three heads: HHHTT, HHTHT, HHTTH, HTHHT, HTHTH, HTTHH, THHHT, THHTH, THTHH and TTHHH. Since $P(H) = P(T) = 0.5$ then the

probability of getting 3 heads and 2 tails (in a given order) is $0.5^3 \times 0.5^2$. Since there are 10 ways to do this, we have $P(X=3) = 10 \times 0.5^3 \times 0.5^2 = 0.3125$.

Example 3.7. A six-sided fair die is rolled 5 times, and let X be 'the number of times I roll a 6'. The probability of getting a 6 is $\frac{1}{6}$ and for each roll I either roll a 6 or not a 6, so we know that $X \sim B(5, \frac{1}{6})$
What is the probability that I roll a 6 exactly once?

Rolling a six exactly once can occur in exactly 5 ways: it can be the 1st, 2nd, 3rd, 4th or 5th roll.
The probability of rolling a 6 is $\frac{1}{6}$ and the probability of not rolling a 6 is $\frac{5}{6}$.

So the probability of rolling one 6 and 4 non-6s (in a given order) is $(\frac{1}{6}) \times (\frac{5}{6})^4$. Since this can happen in 5 different ways we have $P(X=1) = 5 \times (\frac{1}{6}) \times (\frac{5}{6})^4 = 0.4019$.

In general, the calculations may be too complicated to do this by hand, so we can use computational tools. Later, we will see other ways to work with binomial distributions that allow us to compute it more easily.

Function: BINOM.DIST() (E/S)

The `=BINOM.DIST()` function will calculate a binomial distribution. It takes four parameters:

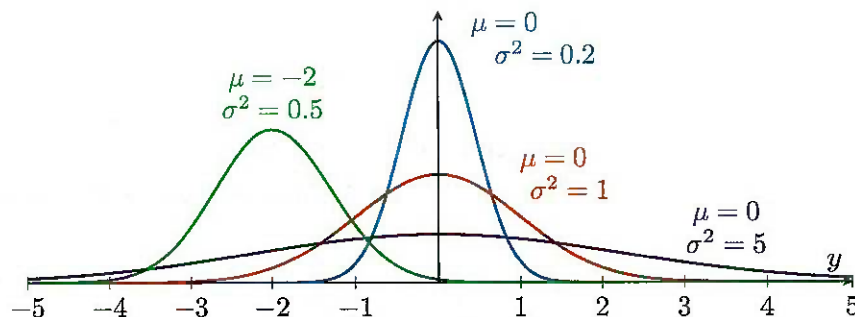
- k , which is the number of successes you wish to find the probability of
- n , the number of independent trials
- p , the probability of success on each trial
- a fourth parameter to specify whether you want the cumulative distribution (TRUE) or not (FALSE); for the probability of 'at most k successes' you want TRUE, and if you set it to FALSE it will tell you the probability of 'exactly k successes'.

3.5 Normal Distribution**Definition 3.8 - Normal Distribution**

The **normal distribution**, sometimes also called a *Gaussian distribution* is a probability distribution for a *continuous random variable* which is commonly used as a model for many real-world distributions. If X is a random variable that follows this distribution, and has a mean of μ and a standard deviation of σ we write:

$$X \sim N(\mu, \sigma^2)$$

The *standard normal distribution* is the version of this distribution with a mean of 0 and a standard deviation of 1, denoted $N(0, 1^2)$.

**Note**

Note that the two numbers in the bracket $N(\mu, \sigma^2)$ are the *Mean and Variance of the distribution*, even though we talk about normal distributions in terms of the standard deviation. It's conventional to leave the expression as σ^2 rather than evaluate it as a number - so for a distribution with a mean of 2 and a standard deviation of 5, we'd write

$$N(2, 5^2)$$

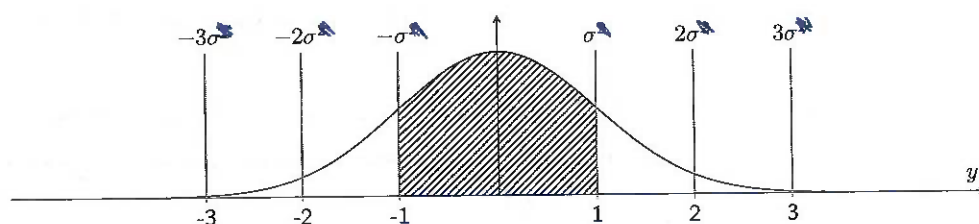
The probability density function for the normal distribution takes the shape of a symmetrical curve, sometimes also called a **bell curve**, which has the equation:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

You don't need to know this equation - but it's worth noting that the curve is entirely specified by **the mean and s.d.** of the distribution.

Varying the mean will move the **centre of the curve** along the horizontal axis, and changing the variance will **stretch / squash** the curve so it's taller/narrower or shorter/wider - to maintain an **area under the curve of 1**.

- The curve is always **left-to-right symmetrical** with the highest point **in the middle**.
- The mean and median of the distribution are **both also in the middle**
- About **$\frac{2}{3}$** of the area under the curve (shaded below) lies within **σ** of the mean.
- About **96% of the area** lies within **$2 \times \sigma$** of the mean.
- Almost all the area (**99.8%**) lies within **$3 \times \sigma$** of the mean.



Since the shape of the curve depends on the **standard deviation**, we can use it as a way to describe points on the curve, in terms of **how far away they are from the mean**.

Definition 3.9 - z-score for a normal distribution

For a given value x the random variable could take, the **z-score** is a way to **locate this value on the normal distribution**.
Given a normal distribution with mean μ and standard deviation σ :

$$z = \frac{x - \mu}{\sigma}$$

Subtracting μ from the value of x tells us **how far away we are from the mean**; dividing by σ **scales this value** so it fits the curve we're looking at. This allows us to find the point on the curve corresponding to the value of x .

Theorem 3.1 - Sum of two normal distributions

Let X and Y be independent random variables that are normally distributed.

Then their sum is also normally distributed.

That is, if $X \sim N(\mu_x, \sigma_x^2)$

$$Y \sim N(\mu_y, \sigma_y^2)$$

and $Z = X + Y$

then

$$Z \sim N(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2)$$

Example 3.8. A baker is weighing ingredients for a cake by eye. Let $X \sim N(200, 15)$ represent the amount of flour weighed out in grams, and $Y \sim N(160, 10)$ represent the amount of sugar. Assume the two are independent.

The total weight of flour and sugar can be expected to follow a normal distribution $X + Y \sim N(360, 25)$

3.5.1 Finding probabilities using a normal distribution

We have described the probability as being related to the area under the curve and there are mathematical techniques for finding the area under a curve between two points; however, when the curve is not a plot of a simple-to-describe curve like $f(x) = x^2$, these techniques are not easy to apply. Instead, the area is pre-calculated for a large number of values, and this information can be looked up using a table or function.

The traditional method (and the one we'll use in the exam) is to consult a table of values like the one given below.

To use this table (and others like it), you need to calculate the z-score of the point x you'd like to know the probability of, for the particular distribution you're working with.

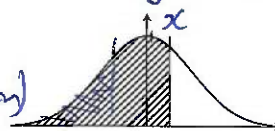
Recall that this can be calculated as $z = \frac{x - \mu}{\sigma}$

If this value is positive, you can look it up in the table: the first digit after the decimal place tells you which row you need, then the second digit after the decimal place is the column. For example, to look up a z-score of 2.14 we'd go to the row corresponding to 2.1 and the column for 0.04 as adding these two numbers together gives 2.14

The number you get from the table will tell you the probability of the value being less than x (which is the area under the curve to the left of x).

Note that when $z = 0$ (no standard deviations away from the mean)

the probability is 0.5 since exactly half the curve lies to each side. If we move away from the mean, the value gets larger (as it includes the left half of the curve, as well as the part between the point you're at and the middle).



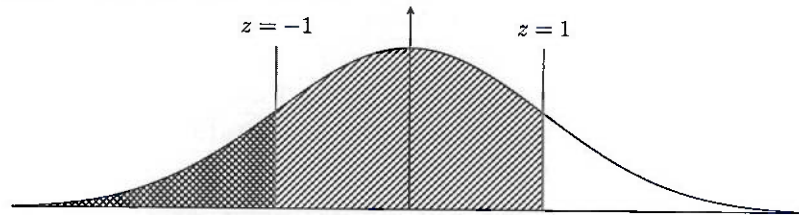
If the z-score is negative, we use the fact that the distribution is symmetric to find the area to the left.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999							

A copy of this table will be included in the exam paper.

Example 3.9. We have calculated the z -score to be -1 . Since this is *negative* we can instead look up the z -score 1 in the table, which will give us *the area to the left of the line at $z=1$* . This is $1 -$ the area we want.

$$\begin{aligned} P(X < -1) &= P(X > 1) \text{ (by symmetry)} \\ &= 1 - P(X \leq 1) \text{ (since the total area is 1)} \\ &= 1 - 0.8413 = 0.1587. \end{aligned}$$



Function: NORM.DIST() (E/S)

The =NORM.DIST() function will calculate a normal distribution. It takes four parameters:

- x , which is the value you want to calculate the probability for
- μ , the mean of the normal distribution you're studying
- σ , the standard deviation of the normal distribution you're studying
- a fourth parameter to specify whether you want the cumulative distribution (TRUE) or not (FALSE); for the area under the curve, you want this to be TRUE, and if you set it to FALSE it will just tell you the height of the curve at that point.

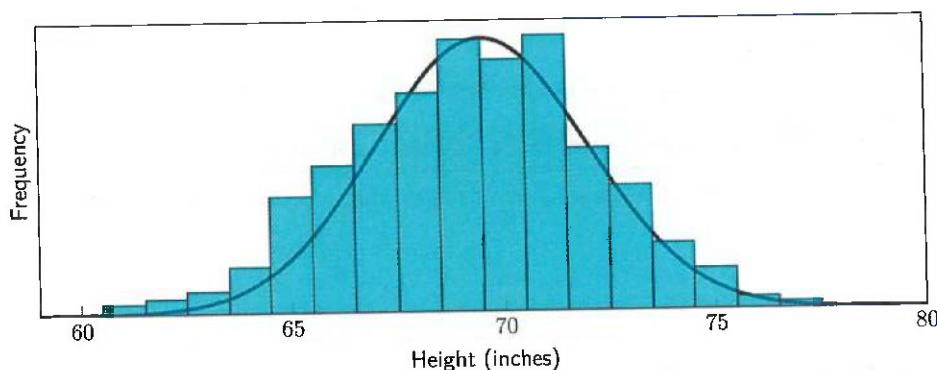
Exercise 3.5.

1. Let X be the distribution of the heights of men from Albania. We assume that X is a normal distribution, with a mean of 174cm and standard deviation of 11 cm. What is the probability that if we spoke to a random Albanian man, he will be taller than 160cm?
2. Let $X \sim N(60.1, 9.392)$.
 - (i) Find $P(X < 58.9)$.
 - (ii) Confirm that $P(X > 57.8)$ is 0.5967.
 - (iii) Find $P(X \geq 70)$.
 - (iv) Confirm that $P(54 \leq X < 58.4)$ is 0.1702.

3.5.2 Applications of the normal distribution

We have been discussing the normal distribution as a *distribution of probabilities*, but this can be thought of as equivalent to *"if I pick a random observation from my dataset, what is the probability its value lies in this range"* and all the same maths applies.

Many real-world datasets are extremely well approximated by the normal distribution - in particular, things which are affected by *many interacting factors*. For example, *the heights of people, birth weights, shoe sizes, and retirement ages of sports players*. fit a normal curve (below), as do



All of these types of data follow a similar shape: most values are *clustered around a mean* and there are *fewer measurements taking extreme values*.

This means the normal distribution can be used to model many real-world datasets. Once it has been established that the distribution is normal, you can use the *mean and SD of the data* to construct a curve which represents a good approximation to the distribution you would expect for the whole population.

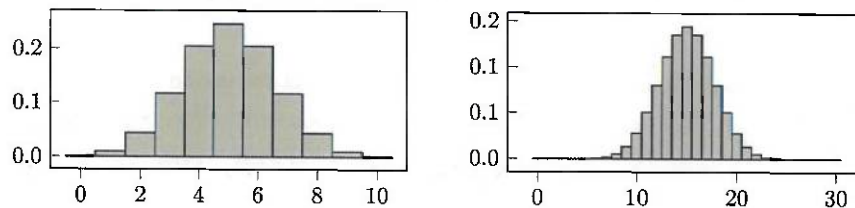
Since the normal distribution is an idealised mathematical curve, it continues on in both directions, *never reaching zero*. This means that it would predict a tiny but non-zero probability of *a person being >4m tall, or <-1m tall*.

This would be such a small probability that in practice you don't need to worry about it, but it's worth remembering that this is a model and not actual data.

Some real-world data does not follow a normal distribution - anything which involves *e.g. exponential growth, or counting instances of something* (where many observations might be zero, with a curve that drops off). It's important to check, by looking at the shape of the data, which distribution is the right one for your model.

3.6 Approximating a Binomial Distribution using a Normal Distribution

The plot below shows *the probability of getting exactly K heads* when a fair coin is flipped 10 times (left), and 30 times (right).



The shape of the binomial distribution *closely mirrors the normal distribution*.
As the number of trials increases, the approximation gets better, and it works best when *p is not close to 0 or 1.*

Definition 3.10 - Normal Approximation to a Binomial Distribution

For $n > 20$, we can approximate a binomial distribution using a normal distribution, with $\mu = np$ and $\sigma^2 = np(1-p)$.

If $X_B \sim B(n, p)$, then $X_N \sim N(np, np(1-p))$ is an approximation for X_B .

Note

The requirement that $n > 20$ is a useful rule of thumb, but larger n give a better approximation.

When we approximate a discrete distribution by a continuous one, under certain conditions we must apply a continuity correction to make the approximation better.

Definition 3.11 - Continuity Correction for the Normal Approximation for Binomials

When approximating $X_B \sim B(n, p)$ using $X_N \sim N(np, np(1-p))$, if the values of np and $n(1-p)$ are both at least 5, we must apply a continuity correction:

$$P(X_B \leq x) \sim P(X_N < x + \frac{1}{2})$$

This correction means that if we are interested in e.g. the probability of the discrete value being ≤ 6 we include the whole of the bar centred at 6, which has width 1.

Example 3.10.

- Consider $X \sim B(5, 0.2)$. Here $n < 20$ so cannot approximate by a normal distribution.
- Consider $X \sim B(25, 0.1)$. We can approximate this by $Y \sim N(np, np(1-p)) = N(2.5, 2.25)$.
Since $np < 5$ we do not need to apply a continuity correction.

$$N(2.5, 2.25)$$

In order to find $P(X < 3)$, we can calculate $z = \frac{x - \mu}{\sigma} = \frac{3 - 2.5}{\sqrt{2.25}} = \frac{0.5}{1.5} = 0.33$

Looking this up in the normal distribution table gives $P(X < 3) = 0.6293$.

- Consider $X \sim B(100, 0.25)$. We can approximate X by $Y \sim N(np, np(1-p))$
 $= N(25, 18.75)$

To find the probability of $19 \leq X \leq 23$, we must first correct for continuity and find

$$P(18.5 < X < 23.5) \quad \text{for } x = 18.5, \quad z = \frac{18.5 - 25}{\sqrt{18.75}} = -1.5$$

$$\text{for } x = 23.5, \quad z = \frac{23.5 - 25}{\sqrt{18.75}} = -0.35$$

The normal distribution table tells us that

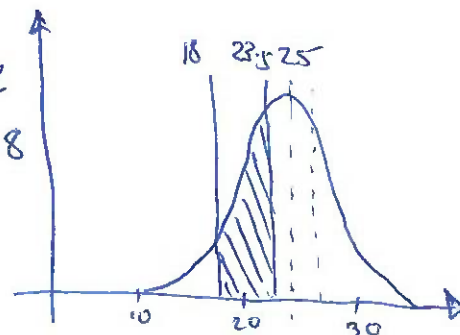
$$P(x < 23.5) = 0.3632$$

Look up 0.35: 0.6368, then $1 - 0.6368 = 0.3632$

$$\text{Similarly } P(x < 18.5) = 1 - 0.9332 = 0.0668$$

Then

$$P(18.5 < x < 23.5) = 0.3632 - 0.0668 = 0.2964.$$



Exercise 3.6.

- Let $X \sim B(60, 0.25)$. Using the normal approximation to the binomial, find $P(X \leq 13)$.
- Let $X \sim B(56, 0.64)$. Using the normal approximation to the binomial, find $P(30 < X < 40)$.

Theorem 3.2 - Linear combinations of a normal distribution

Let X be a normal random variable with mean μ and variance σ^2 . Let a and b be two constants (with $b \neq 0$). Then the random variable Y defined by $Y = a + bX$ has a normal distribution with mean $a + b\mu$ and variance $b^2\sigma^2$ (standard deviation is $b\sigma$).

Example 3.11. Let $X \sim N(10, 2 \cdot 2^2)$ be a normal random variable representing the weight of turkeys. The ^{time} needed ~~time~~ to cook a turkey is given by one recipe as 80 minutes for a 4kg turkey, plus an extra 25 minutes per additional Kg. This means the number of minutes is $80 + 25(X - 4)$
 $= 80 + 25X - 100 = 25X - 20$.

Then the number of minutes you might need to cook a turkey can be modelled by the random variable

$$Y = 25X - 20, \text{ for which } a = -20, b = 25 \quad \text{so: } Y \sim N(25\mu - 20, 25^2 \times 2 \cdot 2^2) \\ = N(230, 55^2)$$